

The Scots College

2001

TRIAL HSC EXAMINATION

Mathematics

Extension 1

General Instructions

- Reading time - 5 minutes
- Working time - 2 hours
- Write using a blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided on page 8
- All necessary working should be shown in every question
- Start each question in a new booklet.

Total Marks: (84)

Weighting: 35% HSC

- Attempt Questions 1 - 7
- All questions are of equal value

Total marks (84)

Attempt Questions 1 – 7

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

- a. Evaluate $\int_0^{2\sqrt{3}} \frac{dx}{4+x^2}$ **2**
- b. Differentiate $\cos^3 x$ **2**
- c. Find the point which divides the line joining (4, 6) and (13, 5) externally in the ratio 4:1 **2**
- d. Write down the equation of the vertical asymptote of $y = \frac{2x}{3x-1}$ **1**
- e. Solve for x : $\frac{3}{x+5} \leq 1$ **2**
- f. Evaluate $\int_0^{\frac{1}{\sqrt{2}}} \frac{2x^3}{\sqrt{1-x^4}} dx$ using the substitution $u = x^4$ **3**

End of Question 1

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. Using all the letters, how many different arrangements can be made from the word MATHEMATICS ? 2
- b. Find all values of θ in the range $0 \leq \theta \leq 2\pi$ for which $\sin \theta + \sqrt{3} \cos \theta = 1$ 4
- c. i. Show that the function $f(x) = 2x^2 + x - 2$ cuts the x axis between $x = 0$ and $x = 1$ 1
- ii. Use the method of halving the interval twice to find an approximation to the root of this equation. 3
- iii. Starting with a value of $x = 0.7$ use Newton's method once to find an approximation to this root correct to 3 decimal places. 2

End of Question 2

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. The region R is bounded by the curve $y = \cos x$, $x = 0$, $x = \frac{\pi}{2}$ and the x -axis.

i. Sketch R

1

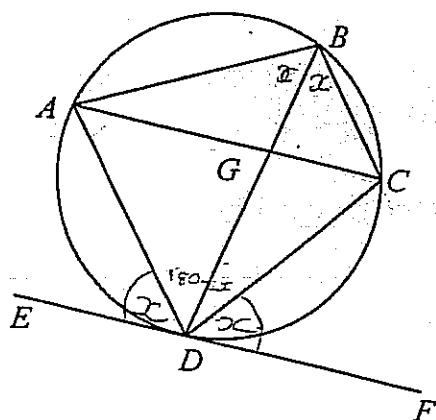
- ii. Find the exact volume of the solid generated when the region R is rotated about the x -axis.

2

- b. If α, β, γ are the roots of the cubic polynomial equation $x^3 + 4x^2 - 6x - 8 = 0$
Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

2

c.



$ABCD$ is a cyclic quadrilateral. EF is a tangent at D . If BD bisects $\angle ABC$, prove that AC is parallel to EF

2

- d. i. By equating coefficients, find the values of A and B in the identity

$$A(2\sin x + \cos x) + B(2\cos x - \sin x) \equiv 7\sin x + 11\cos x$$

2

- ii. Hence show that $\int_0^{\frac{\pi}{2}} \frac{7\sin x + 11\cos x}{2\sin x + \cos x} dx = \frac{5\pi}{2} + \ln 8$

3

End of Question 3

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. P is a variable point on the parabola $x^2 = 8y$ with parameter p . The normal at P cuts the y axis at A and R is the midpoint of AP .
- i. Show that the normal at P has equation $x + py = 4p + 2p^3$ 2
- ii. Show that R has coordinates $(2p, 2p^2 + 2)$ 2
- iii. Show that the locus of R is a parabola and show that the vertex of this parabola is the focus of the parabola $x^2 = 8y$. 3
- b. i. Evaluate $\int_1^3 \frac{dx}{x}$ 1
- ii. Use Simpson's rule with 3 function values to approximate $\int_1^3 \frac{dx}{x}$ 2
- iii. Use your results to parts i and ii to obtain an approximation for e . Give your answer correct to 3 decimal places. 2

End of Question 4

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. Prove by induction that, for all integers $n \geq 1$,

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1} \quad 3$$

- b. i. Find the domain over which the function $y = x^2 + 6x$ is monotonic increasing. 2

- ii. Find the inverse function over this restricted domain, and sketch a graph of this inverse function clearly showing its domain and range. 3

iii. Evaluate $\cos \left[\tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) \right]$ 1

iv. Sketch the graph of $y = 3 \sin^{-1} \left(\frac{x}{2} - 1 \right)$ 3

End of Question 5

Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. When the temperature T of a certain body is 65°C it is cooling at the rate of 1°C per minute.

Assuming Newton's law of cooling: $\frac{dT}{dt} = -k(T - S)$ where

T is the temperature of the body at time t minutes

S is the temperature of the surrounding medium

k is a constant

- i. Verify that $T = S + Ae^{-kt}$ is a solution of the given differential equation, where A is a constant. 2
- ii. Determine the value of k given that S , which is constant, is 15°C . 2
- iii. Find T when $t = 20$ minutes, giving your answer to the nearest degree 2
- iv. How long will it take for the temperature of the body to fall to 35°C ? 2

- b. The acceleration of a particle P , moving along a straight line has an acceleration given by

$$\frac{d^2x}{dt^2} = -4\left(x + \frac{16}{x^3}\right)$$

- i. Given that P is initially at rest at the point $x = 2$ m, show that the velocity v m/s at any time is given by

3

$$v^2 = 4\left(\frac{16 - x^4}{x^2}\right)$$

- ii. Hence, or otherwise, show that when P is halfway to the origin, the speed is given by $2\sqrt{15}$ m/s

1

End of Question 6

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

- a. An arrow is fired horizontally at 60ms^{-1} from the top of a 20m high wall. Taking $g = 10\text{ms}^{-2}$
- i. Show, using calculus, that the horizontal and vertical components of the arrows motion are given by
- $$x = 60t$$
- $$y = -5t^2 + 20$$
- ii. Find the time taken for the arrow to hit the ground.
- iii. Find the distance that the point of impact will be from the base of the wall.
- iv. Find the angle with which the arrow will strike the ground.
- b. A squad of 8 is chosen at random from 3 baseball teams A, B and C with 10 players in each team.
- i. If 5 of the squad are chosen from the A team, 2 from the B team and 1 is chosen from the C team, in how many ways can the squad be formed?
- ii. Find the probability that Joe from the B team and Fred from the A team will be chosen.

End of paper

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2}) + C$$

NOTE : $\ln x \equiv \log_e x, \quad x > 0$

Question 1

$$1. \int_0^{2\sqrt{3}} \frac{dx}{4+x^2} = \left[\frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) \right]_0^{2\sqrt{3}}$$

$$= \frac{1}{2} \tan^{-1}\sqrt{3} - 0$$

$$= \frac{\pi}{6} //$$

[2]

$$b. \frac{d}{dx} (\cos^3 x) = 3 \sin x \cdot \cos^2 x$$

[2]

$$c. \begin{array}{ll} x_1 = 4 & x_2 = 13 \\ y_1 = 6 & y_2 = 5 \\ m = 4 & n = -1 \end{array}$$

$$x = \frac{(-1)(4) + (4)(13)}{4-1}$$

$$= 16$$

$$y = \frac{(-1)(6) + (4)(5)}{4-1}$$

$$= \frac{14}{3}$$

∴ the point required is $(16, \frac{14}{3})$

[2]

$$d. y = \frac{2x}{3x-1}$$

$x = \frac{1}{3}$ is the equation of the vertical asymptote.

[1]

e

$$x \geq -2 \text{ or } x \leq -5$$

$$f. \int_0^{\frac{1}{\sqrt{2}}} \frac{2x^3}{\sqrt{1-x^4}} \cdot dx \quad u = x^4$$

$$\frac{du}{dx} = 4x^3$$

$$= \frac{1}{2} \int_0^{\frac{1}{4}} \frac{1}{\sqrt{1-u}} \cdot du \quad u_1 = \left(\frac{1}{\sqrt{2}}\right)^4$$

$$= \frac{1}{2} \int_0^{\frac{1}{4}} (1-u)^{-\frac{1}{2}} \cdot du \quad u_2 = 0^4$$

$$= 0$$

$$= \frac{1}{2} \left[-\frac{2}{3} (1-u)^{\frac{3}{2}} \right]_0^{\frac{1}{4}}$$

$$= \frac{1}{2} \left(-\frac{\sqrt{3}}{4} \right) - \frac{1}{2} \left(-\frac{2}{3} \right)$$

$$= -\frac{\sqrt{3}}{8} + \frac{1}{3} //$$

[3]

Question 2

$$a. \# \text{ arrangements} = \frac{9!}{2!2!2!}$$

$$= 45360 //$$

[2]

$$t = \frac{\sqrt{3}-1}{-\sqrt{3}-1}$$

$$\tan \frac{\theta}{2} = \frac{\sqrt{3}-1}{-\sqrt{3}-1} \quad 0 \leq \frac{\theta}{2} \leq \pi$$

$$\frac{\theta}{2} = \frac{11\pi}{12}$$

$$\theta = \frac{11\pi}{6}$$

$$\therefore \theta = \frac{\pi}{2}, \frac{11\pi}{6} //$$

[4]

$$\text{let } t = \tan \frac{\theta}{2}$$

$$\Rightarrow \frac{2t}{1+t^2} + \sqrt{3} \left(\frac{1-t^2}{1+t^2} \right) = 1$$

$$2t + \sqrt{3} - \sqrt{3} \cdot t^2 = 1 + t^2$$

$$(-\sqrt{3}-1)t^2 + 2t + (\sqrt{3}-1) = 0$$

$$t = \frac{-2 \pm \sqrt{4 + (\sqrt{3}+1)(\sqrt{3}-1)}}{-2(\sqrt{3}+1)}$$

$$= \frac{-2 \pm \sqrt{12}}{-2(\sqrt{3}+1)}$$

$$= 1 \text{ or } \frac{\sqrt{3}-1}{-\sqrt{3}-1}$$

$$\underline{t=1}: \tan \frac{\theta}{2} = 1 \quad 0 \leq \frac{\theta}{2} \leq \pi$$

$$\frac{\theta}{2} = \frac{\pi}{4}$$

$$\theta = \frac{\pi}{2} //$$

c.

$$i. f(x) = 2x^2 + x - 2$$

$$f(0) = 2(0)^2 + 0 - 2$$

$$= -2$$

$$f(1) = 2(1)^2 + 1 - 2$$

$$= 1$$

$$\text{so } f(0) < 0 \text{ and } f(1) > 0$$

$\therefore f(x)$ must cut the x -axis between $x=0$ and $x=1$.

[1]

$$\begin{aligned} \text{ii } f\left(\frac{0+1}{2}\right) &= f(0.5) \\ &= 2(0.5)^2 + 0.5 - 2 \\ &= -1 \end{aligned}$$

$\therefore$ root lies between $x=0.5$ and $x=1$

$$\therefore \text{ choose } x = \frac{0.5+1}{2} = 0.75$$

$$\begin{aligned} f(0.75) &= 2(0.75)^2 + 0.75 - 2 \\ &= -0.125 \end{aligned}$$

$\therefore$ root lies between

$$x=0.75 \text{ and } x=1$$

$x=0.75$ is our approximation.

[3]

$$\text{ii } x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f(x) = 2x^2 + x - 2$$

$$f'(x) = 4x + 1$$

choosing $x_1 = 0.7 \dots$

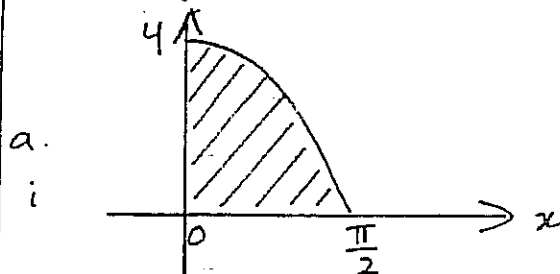
$$x_2 = 0.7 - \frac{f(0.7)}{f'(0.7)}$$

$$= 0.7 - \left(\frac{-0.32}{3.8} \right)$$

$$= 0.784 \quad \parallel \quad (3 \text{ d.p.'s})$$

[2]

Question 3:



$$\begin{aligned} \text{ii } V &= \pi \int_0^{\pi/2} \cos^2 x \cdot dx \\ &= \frac{\pi}{2} \int_0^{\pi/2} (\cos 2x + 1) \cdot dx \\ &= \frac{\pi}{2} \left[\frac{1}{2} \sin 2x + x \right]_0^{\pi/2} \\ &= \frac{\pi}{2} \left(\frac{\pi}{2} \right) - \frac{\pi}{2} (0) \\ &= \frac{\pi}{4} \text{ sq. units. } \parallel \end{aligned}$$

$$\text{b. } x^3 + 4x^2 - 6x - 8 = 0$$

let the roots be α, β, γ

$$\alpha + \beta + \gamma = -4$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = -6$$

$$\alpha\beta\gamma = 8$$

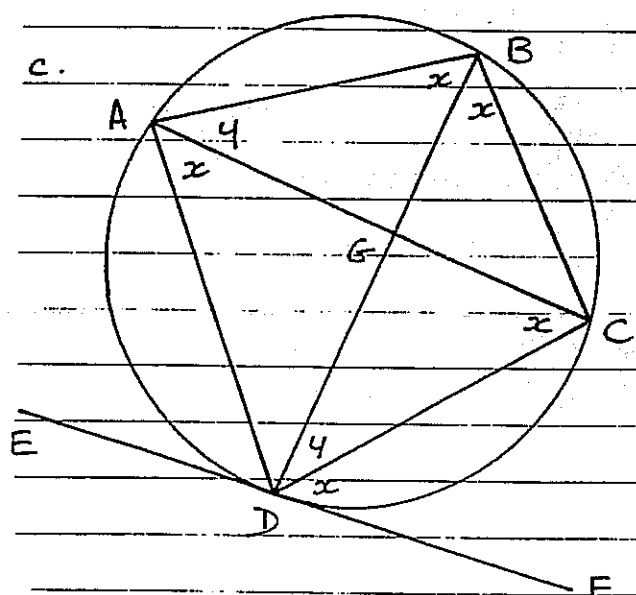
$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{-6}{8}$$

$$= -\frac{3}{4} //$$

[2]



$$\text{let } \angle DBC = x$$

$$\angle DAC = x \text{ (}\angle\text{'s on same arc)}$$

$$\angle ABD = x \text{ (BD bisects } \angle ABC)$$

$$\angle ACD = x \text{ (}\angle\text{'s on same arc)}$$

$$\text{let } \angle BAC = y$$

$$\angle BDC = y \text{ (}\angle\text{'s on same arc)}$$

$$\text{and } \angle BDF = x+y \text{ (}\angle \text{ in alt. seg.)}$$

$$= \angle BAD$$

$$\therefore \angle CDF = \angle BDF - \angle BDC$$

$$= (x+y) - y$$

$$= x$$

$$= \angle ACD$$

$\therefore$ alternate $\angle$'s $\angle CDF$ and $\angle ACD$ are equal

$$\therefore AC \parallel EF //$$

[2]

d.

i

$$A(2\sin x + \cos x) + B(2\cos x - \sin x)$$

$$\equiv 7\sin x + 11\cos x$$

$$2A\sin x + A\cos x + 2B\cos x - B\sin x$$

$$\equiv 7\sin x + 11\cos x$$

$$(2A - B)\sin x + (A + 2B)\cos x$$

$$\equiv 7\sin x + 11\cos x$$

$$\Rightarrow 2A - B = 7 \quad \text{--- (1)}$$

$$A + 2B = 11 \quad \text{--- (2)}$$

from (1)

$$B = 2A - 7$$

subbing into (2) ...

$$A + 2(2A - 7) = 11$$

$$A + 4A - 14 = 11$$

$$5A = 25$$

$$A = 5$$

$$2(5) - B = 7$$

$$B = 3$$

$$\therefore A = 5, B = 3 //$$

[2]

$$\int_0^{\pi/2} \frac{7\sin x + 11\cos x}{2\sin x + \cos x} \cdot dx$$

$$= \int_0^{\pi/2} \frac{5(2\sin x + \cos x) + 3(2\cos x - \sin x)}{2\sin x + \cos x} \cdot dx$$

$$= \int_0^{\pi/2} 5 + 3 \frac{2\cos x - \sin x}{2\sin x + \cos x} \cdot dx$$

$$= \left[5x + 3 \ln(2\sin x + \cos x) \right]_0^{\pi/2}$$

$$= \left(\frac{5\pi}{2} + 3 \ln 2 \right) - (0 + 3 \ln 1)$$

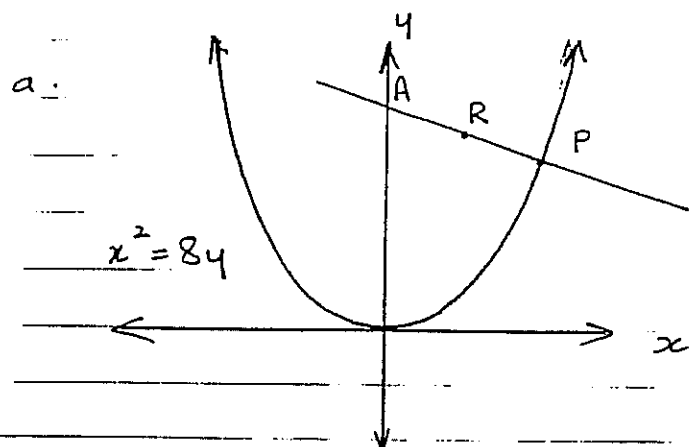
$$\frac{5\pi}{2} + 3 \ln 2$$

$$\frac{5\pi}{2} + \ln 2^3$$

$$\frac{5\pi}{2} + \ln 8 //$$

[3]

Question 4:



i coords of P : $(2ap, ap^2)$
 where $a = 2$
 $\therefore$ P is $(4p, 2p^2)$.

gradient of normal = $-\frac{1}{P}$

$$y - 2p^2 = -\frac{1}{P}(x - 4p)$$

$$py - 2p^3 = -x + 4p$$

$\therefore x + py = 4p + 2p^3$ is the equation of the normal at P.

[2]

ii when $x = 0$, $y = 4 + 2p^2$
 $\therefore$ A is $(0, 4 + 2p^2)$

coords of R : $\left(\frac{4p}{2}, \frac{4 + 2p^2}{2}\right)$
 $= (2p, 2 + p^2)$

[2]

iii

Coords of R:

$$x = 2p \quad \text{--- (1)}$$

$$y = 2 + 2p^2 \quad \text{--- (2)}$$

from (1) $p = \frac{x}{2}$

subbing into (2) gives

$$y = 2 + 2\left(\frac{x}{2}\right)^2$$

$$= 2 + \frac{2x^2}{4}$$

$$= 2 + \frac{x^2}{2}$$

$x^2 = 2(y - 2)$ is the locus of R.

This is a parabola with vertex $(0, 2)$.

now focus of $x^2 = 8y$ is $(0, a)$ where $a = 2$ ie $(0, 2)$ which is the same as the vertex of $x^2 = 2(y - 2)$ //

$$\text{b. i } \int_1^3 \frac{dx}{x}$$

$$= \left[\ln x \right]_1^3$$

$$= \ln 3 - \ln 1$$

$$= \ln 3 //$$

[1]

$$\therefore 3 \doteq e^{\frac{10}{9}}$$

$$\Rightarrow e \doteq 3^{\frac{9}{10}}$$

$$\doteq 2.688 \text{ (3 d.p.'s)}$$

[2]

Question 5:

a. Prove...

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)}$$

$$= \frac{n}{n+1} \text{ for } n \geq 1$$

let $n=1$:

$$\text{LHS} = \frac{1}{1 \times 2} = \frac{1}{2}$$

$$\text{RHS} = \frac{1}{1+1} = \frac{1}{2} = \text{LHS}$$

$\therefore$ true for $n=1 //$

Assume true for $n=k$

$$\text{i.e. } \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$$

$$\text{ii } \int_1^3 \frac{dx}{x} \doteq \frac{1}{3} \{ f(1) + 4f(2) + f(3) \}$$

$$f(x) = \frac{1}{x}$$

$$f(1) = 1 ; f(2) = \frac{1}{2} ; f(3) = \frac{1}{3}$$

$$\therefore \int_1^3 \frac{dx}{x} \doteq \frac{1}{3} \left\{ 1 + 2 + \frac{1}{3} \right\}$$

$$\doteq \frac{10}{9}$$

[2]

iii from i and ii

$$\ln 3 \doteq \frac{10}{9}$$

Prove true for $n=k+1$

$$HS = \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{k(k+2) + 1}{(k+1)(k+2)}$$

$$= \frac{k^2 + 2k + 1}{(k+1)(k+2)}$$

$$= \frac{(k+1)^2}{(k+1)(k+2)}$$

$$= \frac{k+1}{k+2}$$

$$RHS = \frac{k+1}{(k+1)+1}$$

$$= \frac{k+1}{k+2}$$

$$= LHS$$

$\therefore$ true for $n=k+1$
 $\therefore$ since true for $n=1$
 then true for $n=2, n=3, \dots$

$\therefore$ true for all $n \geq 1$ // [3]

b.

$$i \quad y = x^2 + 6x$$

$$- \quad \frac{dy}{dx} = 2x + 6$$

for monotonic increasing... $\frac{dy}{dx} > 0$

$$2x + 6 > 0$$

$$2x > -6$$

$$x > -3$$

the function is monotonic increasing when $x > -3$

[2]

$$ii \quad \text{let } x = y^2 + 6y$$

$$x + 9 = y^2 + 6y + 9$$

$$= (y + 3)^2$$

$$y + 3 = \pm \sqrt{x + 9}$$

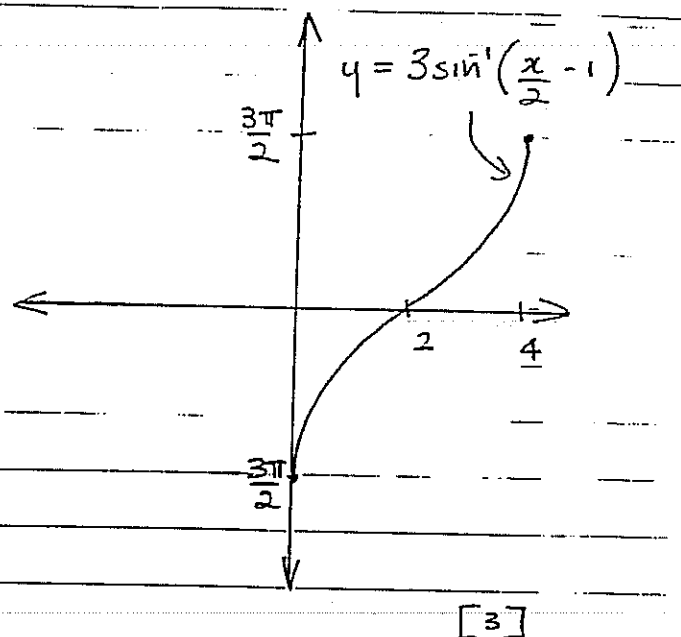
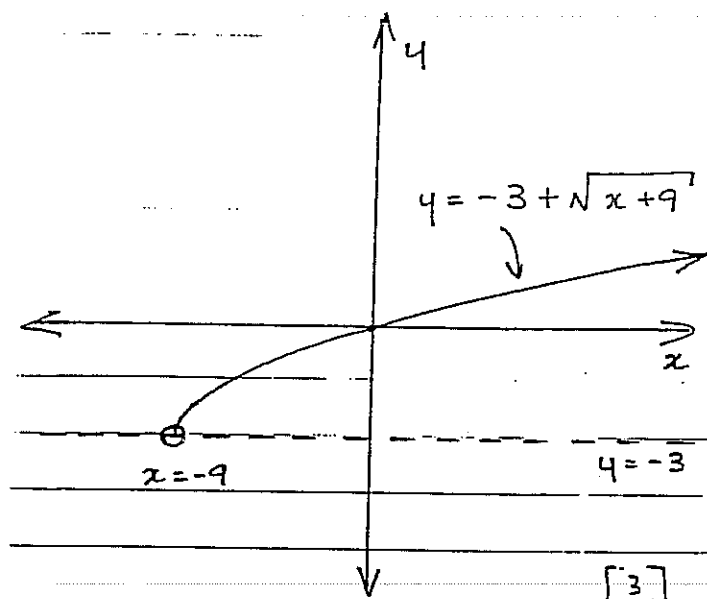
$$y = -3 \pm \sqrt{x + 9}$$

but the range will be $y > -3$

$\therefore y = -3 + \sqrt{x + 9}$ //
 is the inverse function.

domain: $x \geq -9$

range: $y > -3$



iii $\cos \left[\tan^{-1} \left(\frac{-1}{\sqrt{3}} \right) \right]$

$$= \cos \left(-\frac{\pi}{6} \right)$$

$$= \cos \left(\frac{\pi}{6} \right)$$

$$= \frac{\sqrt{3}}{2} //$$

$[1]$

iv. $y = 3\sin^{-1} \left(\frac{x}{2} - 1 \right)$

$$-1 \leq \frac{x}{2} - 1 \leq 1$$

$$-2 \leq x - 2 \leq 2$$

$$0 \leq x \leq 4$$

domain : $0 \leq x \leq 4$

range : $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

Question 6:

i $\frac{dT}{dt} = -k(T-S)$

let $T = S + Ae^{-kt}$

$$\frac{dT}{dt} = -Ake^{-kt}$$

$$-k(T-S) = -k(S + Ae^{-kt} - S)$$

$$= -Ake^{-kt}$$

$$= \frac{dT}{dt}$$

$\therefore T = S + Ae^{-kt}$ is a solu. of the differential equation.

$[2]$

ii Initial conditions:

$$t=0, S=15, \frac{dT}{dt} = -1, T=65$$

$$T = S + Ae^{-kt}$$

$$65 = 15 + A \cdot 1$$

$$\therefore A = 50$$

$$\frac{dT}{dt} = -k(T-S)$$

$$-1 = -k(65-15)$$

$$\therefore k = \frac{1}{50} //$$

[2]

$$\text{iii } T = 15 + 50e^{-\frac{t}{50}}$$
$$= 15 + 50e^{-\frac{20}{50}}$$

$$= 49^\circ \text{ (to nearest degree)}$$

[2]

$$\text{iv } 35 = 15 + 50e^{-\frac{t}{50}}$$
$$20 = 50e^{-\frac{t}{50}}$$

$$e^{-\frac{t}{50}} = 0.4$$

$$-\frac{t}{50} = \ln(0.4)$$

$$t = -50 \ln(0.4)$$

$$= 45.8 \text{ minutes.} //$$

[2]

b.

i

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -4 \left(x + 16x^{-3} \right)$$

$$\therefore \frac{1}{2} v^2 = \int -4x - 64x^{-3} \cdot dx$$

$$= -2x^2 + 32x^{-2} + C$$

$$\therefore v^2 = -4x^2 + \frac{64}{x^2} + D$$

$$\text{now } v=0 \text{ when } x=2$$

$$\therefore 0 = -16 + 16 + D$$

$$\therefore D = 0$$

$$\text{so } v^2 = \frac{64}{x^2} - 4x^2$$

$$= 4 \left(\frac{16}{x^2} - x^2 \right)$$

$$= 4 \left(\frac{16 - x^4}{x^2} \right) //$$

[3]

ii when P is halfway to the origin $x=1$.

$$v^2 = 4 \left(\frac{16-1}{1} \right)$$

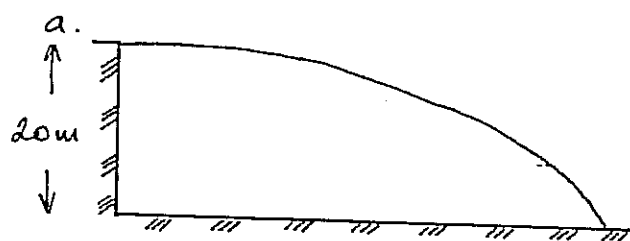
$$= 4 \times 15$$

$$\therefore v = \pm 2 \times \sqrt{15}$$

hence speed is $2\sqrt{15} \text{ m/s.}$

[1]

Question 7:



i horizontal motion:

$$\ddot{x} = 0$$

$$\dot{x} = \int 0 \cdot dt$$

$$= C$$

since $\dot{x} = 60$ when $t = 0$

$$C = 60$$

$\therefore \dot{x} = 60$ and is constant.

$$x = \int 60 \cdot dt$$

$$= 60t + D$$

$$x = 0 \text{ when } t = 0$$

$$\therefore D = 0$$

$$\therefore x = 60t //$$

vertical motion:

$$\ddot{y} = -10$$

$$\dot{y} = \int -10 \cdot dt$$

$$= -10t + E$$

$$= 0 \text{ when } t = 0$$

$$\therefore E = 0$$

$$y = \int -10t \cdot dt$$

$$= -5t^2 + F$$

$$= 20 \text{ when } t = 0$$

$$\therefore F = 20$$

$$\therefore y = -5t^2 + 20 //$$

[3]

$$\text{ii } 0 = -5t^2 + 20$$

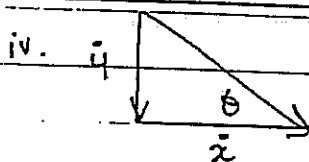
$$5t^2 = 20$$

$$t^2 = 4$$

$$t = 2 \text{ s.} //$$

[2]

$$\text{iii when } t = 2, x = 120 \text{ m. [1]}$$



$$\tan \theta = \dot{y} / \dot{x}$$

$$\text{when } t = 2, \dot{y} = -20$$

$$\dot{x} = 60$$

$$\therefore \tan \theta = -1/3$$

$$\theta = \tan^{-1}(1/3)$$

$$= 18.4^\circ \text{ with the horizontal.}$$

[2]

b.

$$\text{i } \# \text{ ways} = \frac{{}^{10}C_5 \times {}^{10}C_2 \times {}^{10}C_1}{5 \times 2 \times 1} = 113400 //$$

[2]

$$\text{ii } \# \text{ ways of choosing Joe and Fred} = {}^9C_4 \times {}^9C_1 \times {}^{10}C_1$$

$$\therefore P(\text{Joe, Fred}) = 11340 / 113400 [2]$$

$$= 0.1 //$$